



# Development of a Fresh and Cold Chain Logistics Information Sharing Platform Based on Blockchain and Tripartite Evolutionary Game Theory

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**Abstract.** In China, challenges have been encountered in managing cold chain information, especially poor information exchange during distribution. This situation has resulted in significant disruptions and losses within the fresh cold chain logistics sector. To tackle the issue of compromised fresh food quality during transportation due to informational gaps, a blockchain-based platform is constructed by this study. The platform is designed to facilitate efficient logistics data exchange for fresh goods. With aims ranging from system design to functional deployment, the initiative seeks to elevate service standards in cold chain logistics. The feasibility of this innovative information-sharing platform is evaluated through a tripartite evolutionary game, involving stakeholders such as fresh food markets, fresh food suppliers, and cold chain logistics companies. The game ensures that transparent and accurate data exchange, efficient product positioning, and reduction in losses of fresh products due to informational silos are achieved.

**Keywords:** Fresh cold chain logistics · Information sharing platform · Blockchain · Tripartite evolutionary game

## 1 Introduction

The findings from the China Federation of Logistics & Purchasing highlight a significant uptick in the demand for fresh food across China, reaching a volume of 350 million tons in 2023, an increase of 6.1%. Additionally, the turnover generated by cold chain logistics services observed a 5.2% rise, amounting to 517 billion yuan. These statistics underscore the swift expansion of the cold chain logistics sector. Despite this growth, the sector confronts enduring challenges [1, 2].

A primary issue is the deficiency in information management practices. A notable number of cold chain enterprises have underinvested in the development of robust information systems. This underinvestment hampers the effective collection, transmission, and dissemination of data along the supply chain. Moreover, the lack of real-time monitoring and traceability systems complicates the timely identification and resolution

of issues, thereby elevating the risks associated with food safety and quality control [3]. Another significant obstacle is the insufficient sharing of information throughout the distribution phase [4]. The level of collaboration and information exchange among producers of fresh food, cold chain logistics providers, and retailers remains suboptimal, leading to informational gaps and delayed responses. These shortcomings inflict significant disruptions and inefficiencies within the fresh food cold chain logistics system. Addressing these challenges is crucial. To fill the relevant research gap mentioned above, this paper introduces prospect theory to the current Chinese cold chain situation and uses it to build a tripartite evolutionary model. It calls for increased investments, improvements in information management, and the enhancement of information-sharing protocols during distribution. Such measures are essential for the betterment of the fresh food cold chain logistics sector and play a pivotal role in driving forward the industry’s development.

2 Platform Design Objectives

Data pertaining to demand from the three involved parties is communicated to the information-sharing platform via their distinct testing technologies. Following this, the platform undertakes the tasks of information matching and dissemination. In instances of issues related to fresh food products, the platform provides traceability services for these items once they reach the market. With the application of cloud computing and artificial intelligence, the platform can provide a more organic delivery solution. This entire process is illustrated in Fig. 1.

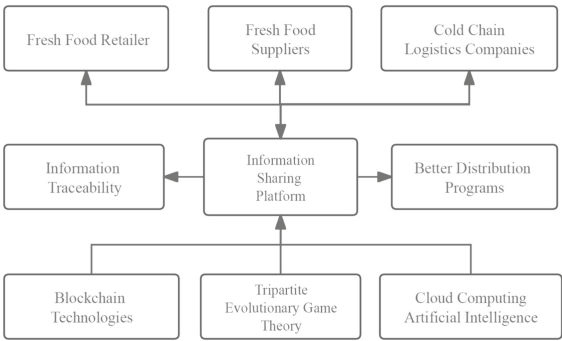


Fig. 1. The framework for designing objectives in an information-sharing platform

Initially, the platform aims to fortify information integrity and refine the mechanism for sharing information during the distribution stages. By leveraging the capabilities of blockchain technology, it ensures that data remains transparent and accessible, while simultaneously maintaining the confidentiality and security of the information through the establishment of a dependable platform [5]. Furthermore, the platform is engineered to support full traceability and real-time surveillance, thereby improving the traceability of freshly harvested agricultural products. This enhancement in traceability and surveillance contributes significantly to

superior quality control and supply chain management efficiency. The application of blockchain technology in creating a traceability system for the logistics of fresh produce allows for the precise recording and monitoring of the origins and transport routes of these goods. This functionality ensures the platform’s ability to provide full traceability, effectively minimizing the risks associated with food safety [6].

Lastly, the platform addresses the challenge of aligning supply with demand through the accurate dissemination of information. It enables the efficient sharing of sales data among the three key parties involved, covering aspects such as supply-demand dynamics, order specifics, transportation status, and delivery timelines. Concurrently, the platform possesses analytical capabilities to forecast transit times for various transportation modes and to formulate plans based on both real-time data and historical transportation records [7–9].

3 Main Framework

3.1 Information Sharing Platform Main Architecture

Exploring the critical challenges associated with the transparency of cold chain information in the present context, this proposed design integrates blockchain technology to facilitate the creation of an information-sharing platform. This platform’s structure is delineated into four distinct segments, as illustrated in Fig. 2.

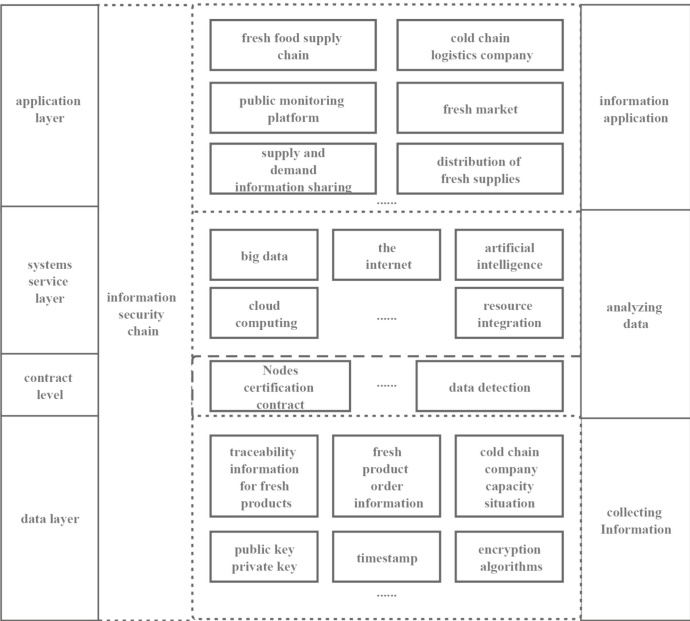


Fig. 2. Principal structure of the fresh cold chain logistics information sharing platform

The data layer encapsulates a variety of pertinent data stored within the blockchain. This encompasses block data, the architecture of the chain, timestamps, hash functions,

Merkle trees, and the public key for random numbers, among others. Primarily, the block data archives traceability details pertinent to the fresh supply chain, order information, and the capacities of cold chain logistics entities. These data may be traced and located in an ordered manner thanks to the features of blockchain, which makes it easier to find the traceability information of new items. Two important components of the system's underlying technology are the smart contract layer and the blockchain data layer.

The contract layer sets forth the fundamental prerequisites necessary for executing the system's diverse functions. It plays a crucial role in instituting smart contracts tailored to these operations. Smart contracts are based on automated and self-governing scripts, algorithms, and intelligent contracts that carry out pertinent tasks based on predetermined criteria without the need for outside involvement. Smart contracts help the system function more swiftly and accurately by preventing human error in contrast to traditional information manipulation. By amalgamating the asymmetrically encrypted blockchain's public and private keys with statistical data, an elevated security and verification standard is established within the blockchain protocol. This methodology ensures the data's integrity and authenticity, thereby bolstering the trust and reliability associated with the blockchain network.

At the system service layer, technical support is extended for harnessing the blockchain functionalities. By scrutinizing fresh product data on the platform, the system is equipped to make informed capacity decisions based on the demand for information on fresh cold chain products. Additionally, it enhances the scheme for matching and processing cold chain order loads, offering intelligent decision-making support [10]. By doing so, obstacles to traditional information data are broken down and the synergistic cooperation of different subjects is realized. Based on the useful data acquired, scientific application management is then implemented to enable the emergency information of fresh items to be utilized in a variety of circumstances, optimizing the information's usefulness and achieving the system's feedback mechanism.

The application layer engages various stakeholders, including fresh food suppliers, cold chain logistics firms, fresh food retailers, public regulatory bodies, platforms for sharing supply and demand information, and mechanisms for emergency order matching. Through the information exchange facilitated by this platform, an efficient matching of fresh food demand and supply is achieved. Suppliers, having instant access to product demand information, are empowered to expedite deliveries, thereby mitigating transportation delays.

### 3.2 Analysis of Information Sharing Services

The conceptualized framework for the cold chain logistics platform's service system, as delineated in this project, is depicted in Fig. 3. This framework encompasses the processes involved in collecting, storing, distributing, analyzing, and utilizing data pertinent to perishable food products.

In examining the myriad of information resources collected, encompassing traceability data for fresh products, tracking information on cold chain logistics capacities, and order data for fresh products, it's clear that these datasets, following collection, undergo a classification and processing phase before being archived on the blockchain. This archival process not only facilitates future reference and traceability but also enhances

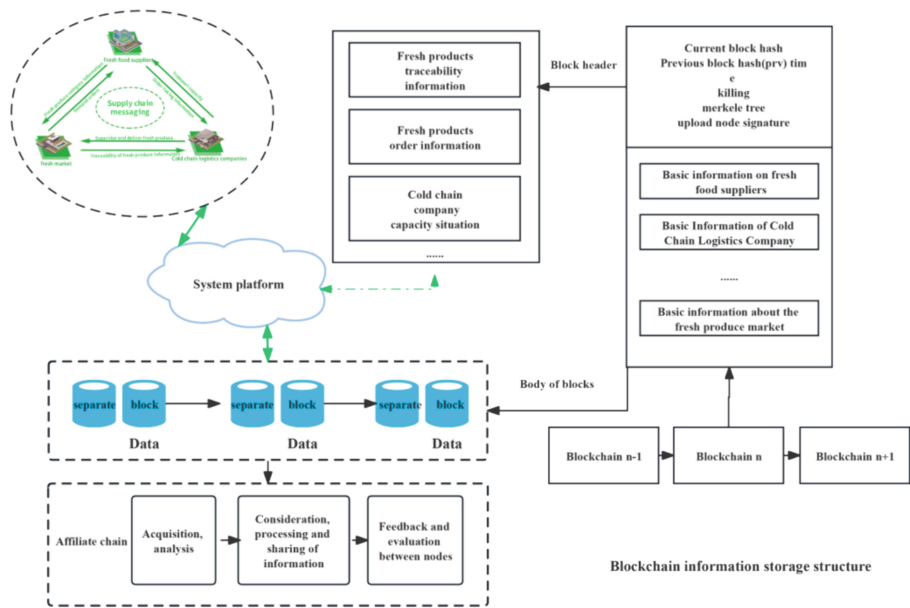


Fig. 3. Components of the new platform for exchanging information on cold chain

the capability for optimal distribution planning and precise allocation of fresh products. The blockchain’s traceability feature extends throughout the entire operational chain, enabling it to support accountability tracing [11]. Should a malfunction occur within any segment of the chain, it can be quickly identified, allowing the responsible parties to be held accountable.

4 Feasibility Analysis of the Information-Sharing Platform

The establishment of an information-sharing platform tailored for cold chain logistics providers, fresh food suppliers, and fresh food retailers serves as a pivotal mechanism for adeptly managing various market dynamics. To assess the viability of this information-sharing platform further, this study will utilize a tripartite evolutionary game model that involves the interaction among these three entities [12, 13].

4.1 Clarify the Stakeholders

Within the confines of this model, the participants of the tripartite evolutionary game are identified as the cold chain logistics providers, fresh food suppliers, and fresh food retailers. The role of the cold chain logistics provider is to ensure the smooth transportation of the entire fresh supply chain, thereby facilitating the flow of fresh goods. The fresh supplier has to procure and dispatch fresh goods, whereas the fresh food retailer’s responsibility lies in merchandising the fresh goods post-receipt from logistics transport.

## 4.2 Model Hypothesis

The fresh produce supply chain is composed of various participants. This includes the cold chain logistics providers, suppliers, and retailers. Through cold chain logistics, fresh produce will be provided to retailers, while retailers will have the right to choose to report problems. Whether to adopt this blockchain-based platform for information sharing is decided upon by three parties together. This study makes the following assumptions to confine the model in order to properly examine this problem.

Hypothesis 1 posits that should the cold chain logistics providers, fresh food suppliers, and fresh food retailers engage with the platform. It is unlikely that any of the parties will cease using the platform shortly.

Hypothesis 2 clarifies that the model does not consider the income generated by the cold chain logistics companies, fresh food suppliers, and fresh food retailers from other avenues. Instead, it focuses exclusively on the revenues generated within this specific food distribution network.

Hypothesis 3 suggests that the reduction in losses post-platform implementation is directly correlated with a decrease in spoilage of perishable goods during transit and storage, a benefit arising from the enhanced exchange of information.

The details and nuances of these parameters are outlined in Table 1.

**Table 1.** Parameter explanation table

| Parameter | Parameter interpretation                                                |
|-----------|-------------------------------------------------------------------------|
| $M$       | Cold chain logistics company's usual income                             |
| $N$       | Regular income of fresh food supplier                                   |
| $R$       | Regular sales in the fresh food market retailer                         |
| $T_1$     | The cost of this platform for cold chain logistics company              |
| $T_2$     | The cost of this platform for fresh food supplier                       |
| $T_3$     | The cost of this platform for fresh food market retailer                |
| $S_1$     | The reduced loss of fresh food supplier after using the platform        |
| $S_2$     | The reduced loss of fresh food market retailer after using the platform |
| $a$       | Goods demand level in the fresh food market retailer                    |
| $b$       | Freshness of the goods (retention time)                                 |
| $p$       | Journey of goods to market                                              |
| $n$       | Freshness per kilometer during cargo transport                          |
| $f$       | Fuel consumption per kilometer during cargo transport                   |

## 4.3 Model Building

The tripartite evolutionary game model is built upon the subject, assumptions, and parameters previously presented. The model is shown in Table 2.

**Table 2.** Tripartite evolutionary game model

| Main body strategy                               |                                                           | Fresh supplier                                                                                         |                                                            |
|--------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|                                                  |                                                           | Use the platform (y)                                                                                   | Not use platform (1y)                                      |
| fresh food retailer use the platform (z)         | Cold chain logistics company use the platform (x)         | MfpT <sub>1</sub><br>NS <sub>1</sub> T <sub>2</sub> bnp<br>RS <sub>2</sub> − T <sub>3</sub> a − b − np | Mfp − T <sub>1</sub><br>N<br>R − T <sub>3</sub> a − b − np |
|                                                  | Cold chain logistics company do not use the platform (1x) | M<br>NS <sub>1</sub> − T <sub>2</sub> − b − np<br>R − T <sub>3</sub> + a − b − np                      | M<br>N<br>R − T <sub>3</sub> a − b − np                    |
| fresh food retailer do not use the platform (1z) | Cold chain logistics company use the platform (x)         | Mfp − T <sub>1</sub><br>NS <sub>1</sub> − T <sub>2</sub> − b − np<br>R                                 | Mfp − T <sub>1</sub> {\break}N<br>R                        |
|                                                  | Cold chain logistics company do not use the platform (1x) | M<br>NS <sub>1</sub> − T <sub>2</sub> − b − np<br>R                                                    | M<br>N<br>R                                                |

Through simplified calculation, the replication dynamic equations of the three parties are obtained as follows:

The replication dynamic equation of the cold chain logistics company:

$$F(x) = x * (x - 1) * (T_1 - f * p) \tag{1}$$

The replication dynamic equation of the fresh supplier:

$$F(y) = y * (y - 1) * (T_2 - S_1 + b + n * p) \tag{2}$$

The replication dynamic equation of the fresh food retailer:

$$F(z) = z * (z - 1) * (T_3 - S_2 * x * y - a + b + n * p) \tag{3}$$

**4.4 Tripartite Equilibrium Point Analysis**

Make  $F(x) = F(y) = F(z) = 0$  to obtain the eigenvalue of the local equilibrium point, as shown in Table 3.

If the eigenvalues of the Jacobi matrix must be negative, the equilibrium point is a point of progressive stability.

The realization of this scenario hinges upon three pivotal conditions. Firstly, the operational cost associated with the platform’s use must be outweighed by the savings accrued from reduced fuel consumption post-adoption. Secondly, the freshness level of the current batch of goods should be comparatively low. Thirdly, the proximity between the goods and the fresh food market retailer must be relatively short. Fulfillment of these conditions ensures that the eigenvalues of the Jacobi matrix, specifically corresponding to the equilibrium point  $E_8(I, I, I)$ , are all negative, signaling the system’s stability. This outcome underscores the robustness and dependability of the system (Table 4).

**Table 3.** Eigenvalues of the equilibrium points of the Jacobi matrix

| Equalization point | Eigenvalue 1 | Eigenvalue 2 | Eigenvalue 3 |
|--------------------|--------------|--------------|--------------|
| $E_1(0,0,0)$       | $fpT_1$      | $aT_3bnp$    | $S_1T_2bnp$  |
| $E_2(1,0,0)$       | $T_1fp$      | $aT_3bnp$    | $S_1T_2bnp$  |
| $E_3(0,1,0)$       | $fpT_1$      | $aT_3bnp$    | $T_2S_1abnp$ |
| $E_4(0,0,1)$       | $fpT_1$      | $T_3abnp$    | $S_1T_2bnp$  |
| $E_5(1,1,0)$       | $T_1fp$      | $T_2S_1bnp$  | $S_2T_3abnp$ |
| $E_6(0,1,1)$       | $T_1fp$      | $T_3a+bnp$   | $S_1T_2bnp$  |
| $E_7(1,0,1)$       | $fpT_1$      | $T_2S_1bnp$  | $T_3S_2abnp$ |
| $E_8(1,1,1)$       | $T_1fp$      | $T_2S_1bnp$  | $T_3S_2abnp$ |

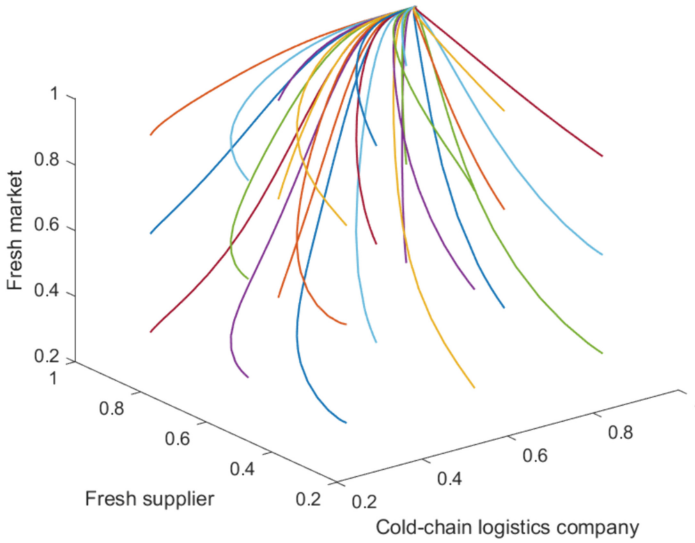
**Table 4.** Equilibrium point stability analysis

| Equalization point | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | Stability    |
|--------------------|-------------|-------------|-------------|--------------|
| $E_1(0,0,0)$       | +           | +           | +           | Saddle point |
| $E_2(1,0,0)$       | −           | +           | +           | Instability  |
| $E_3(0,1,0)$       | +           | −           | +           | Instability  |
| $E_4(0,0,1)$       | +           | +           | −           | Instability  |
| $E_5(1,1,0)$       | −           | −           | +           | Instability  |
| $E_6(0,1,1)$       | −           | −           | −           | Instability  |
| $E_7(1,0,1)$       | +           | −           | −           | Instability  |
| $E_8(1,1,1)$       | −           | −           | −           | ESS          |

In the simulation data,  $T_1 = 10, T_2 = 10, T_3 = 10, S_1 = 50, S_2 = 50, a = 15, b = 10, p = 10, f = 2, n = 1$ .

As shown in Fig. 4, the evolutionary model allows for the deduction that cold chain logistics companies, fresh food suppliers, and fresh food retailers are all poised to achieve a stable equilibrium through engagement with the information-sharing platform. This evidences the platform’s significant utility and effectiveness in facilitating real-world logistical operations.





**Fig. 4.** 3D simulation of the  $E(I, I, I)$  equilibrium point

## 5 Conclusion

This paper intends to create a platform that uses blockchain technology to exchange information about fresh food logistics. Furthermore, the tripartite evolutionary game is employed to assess the feasibility of this proposition, suggesting the collaboration of the three parties in creating a framework for sharing information.

- 1) Through the application of blockchain technology, the platform addresses information asymmetry, ensuring seamless data exchange among logistics providers, suppliers, and markets. The strategy aims to minimize losses associated with informational gaps during transportation, optimize resource allocation, and develop a cooperative supply chain ecosystem.
- 2) Moreover, traceability issues within fresh produce logistics are tackled by the platform through blockchain's decentralized and transparent nature. Data across the cold chain is meticulously recorded and managed, ensuring comprehensive traceability for accurately tracking produce origins and transportation routes, thereby reducing food safety risks.
- 3) In addition, the platform meets challenges related to synchronizing fresh produce delivery with harvest times and balancing delivery speed with costs. By integrating AI, big data, the internet, and cloud computing, it analyzes data such as fresh goods details and transportation expenses. This analysis aids in coordinating delivery and harvest schedules, optimizing the balance between timeliness and cost in cold chain logistics for fresh produce, and ultimately reducing both product losses and logistics costs.

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